

SANTANU DAS

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SUMMARY

Backend engineer and technical lead with 8 years designing distributed data systems in Java. At CertifyOS, set the architecture for and built the company's provider Master Data Management platform from zero; it is now the system of record for provider data across 4 downstream products. Experience spans healthcare data, fintech rewards infrastructure (Paytm), and consumer lending (Credit Saison). Operates across the full arc: problem framing, architecture, delivery through a team, and production ownership.

EXPERIENCE

Team Lead, CertifyOS

Oct 2024 - Present

- **Provider MDM platform (0 to 1):** Owned architecture and delivery end to end: multi-source ingestion, entity resolution, survivorship, and golden-record APIs on Java/Quarkus, Cloud Spanner, and Pub/Sub. It replaced legacy attribution engine and manual reconciliation, giving 4 downstream services (Rosters, Credentialing, Monitoring and platform console) one consistent provider record.
- **Entity resolution engine:** Designed matching that combines deterministic keys (NPI, license, TIN) with probabilistic similarity scoring on demographics and addresses, using blocking to keep candidate generation tractable at scale. Reached 96% precision and cut manual data-steward review by 80%.
- **Survivorship service:** Implemented rule-driven survivorship (source trust hierarchy, recency, completeness) to compute golden records with field-level lineage, making every attribute auditable back to its originating source.
- **Cloud Spanner data layer:** Modelled the provider data graph on Google Cloud Spanner using interleaved tables, secondary indexes, and bounded-staleness reads; eliminated hotspots and tuned transaction patterns to serve **800 QPS** at **200 ms p99**.
- **Event-driven ingestion:** Built asynchronous ingestion and reprocessing on Pub/Sub with idempotent consumers and replay. This enabled full re-matching of 1M+ records from 14 external sources with zero downtime whenever match rules change.
- **Technical leadership:** Set technical direction for the MDM domain. Authored design docs and ran architecture reviews, split the roadmap into parallel workstreams across a team of 4, and mentored engineers on Spanner data modelling and distributed-systems design. Outcome: 3 engineers now independently own services.

Senior Software Engineer, Paytm

Sep 2022 - Oct 2024

- **Traffic replay framework:** Built a traffic replay module for Spring Boot microservices that mirrors production traffic to canary hosts with automated response diffing and metric publishing, at under 3% CPU overhead; enabled release validation against real traffic before rollout.
- **Anomaly detection and alerting:** Designed z-score based Prometheus queries to detect abnormal and fraudulent campaign budget distribution, saving an estimated **USD 1200 per month**; owned the alert notification service that pages campaign owners automatically during incidents, removing hours of manual stakeholder follow-up.
- **Reliability and performance:** Led hardening of cashback services with query timeouts, caching, and connection tuning, plus capacity sizing via Locust load tests and CPU profiling. Refactored cache-refresh jobs using the observer pattern, cutting redundant database calls by 20% and eliminating cascading failures during peak campaigns.

Software Engineer 2, Credit Saison India

Mar 2021 - Sep 2022

- **Loan pre-approval architecture:** Architected and shipped the consumer-lending pre-approval flow powering CRED's loan journey, returning approval decisions in under 10 seconds; deployed the Java service on AWS through CloudFormation and CodePipeline templates driven by shell automation.
- **Read-replica traffic routing:** Implemented read-only transaction routing in Spring Boot over AWS RDS Aurora using a dedicated PostgreSQL datasource and HikariCP pooling, shifting read traffic to replicas and reducing primary database load by 50%.
- **Co-lending workflow engine:** Designed BPMN workflows in Activiti for co-lending partners; indexed lifecycle events in Elasticsearch to serve on-demand loan status lookups and partner callbacks.

Associate Consultant, Oracle Financial Services Software

Aug 2018 - Feb 2021

- Implemented core banking logic in PL/SQL across multiple client deployments. Migrated hot tables to Oracle In-Memory, improving transaction processing 20 to 30% with no application changes.

EDUCATION

Bachelor of Technology, Computer Science and Engineering

Jul 2014 - Jun 2018

National Institute of Technology, Agartala

SKILLS

Languages & Frameworks: Java, Spring Boot, Golang, JavaScript, Python

Data: Oracle DB, PostgreSQL, MySQL, MongoDB, Spanner

Cloud: GCP (Cloud Spanner, Cloud Run, Pub/Sub); AWS (RDS Aurora, CloudFormation, CodePipeline)

Infrastructure: Docker, Kubernetes, Jenkins, Prometheus, Grafana, Apache Spark, Locust

Practices: Distributed Systems, Microservices, Master Data Management, Entity Resolution, System Design, CI/CD